ESTIMATION OF ROOFTOP SOLAR PV SYSTEM (1 - 20 kW)

Date: 22-Sep-2026

Site: Case A - Sufficient Area Customer | Location: Bengaluru, Karnataka | Grid-Tied Solar Engineering Dossier

PLANT CAPACITY	ANNUAL YIELD	CAPITAL COST	PAYBACK PERIOD
550W Mono	Specific Yield	12-Item BOM	25Y Lifecycle
15.00 kW	**24,417 kWh**	**Rs. 656,083**	**4Y 5M**
15.40 kWp Installed	CUF 18.1% | PR 75%	Rs. 6.56 Lakhs	Profit: Rs. 72.42L

[SITE] Customer & Site Engineering Parameters

Verified project configuration, contact details, and sizing pathway

Company / Customer Name	Case A - Sufficient Area Customer
Contact Mobile / Phone	+91 98888 11111
Location / Address	Bengaluru, Karnataka
Contact Email Address	case_a@voltroof.com
Design Path Selected	Rooftop Area Analysis
Approved Sanctioned Load (CD)	15 kW (KERC / ESCOM Tariff Order)
Rooftop Area Verification	Sufficient Rooftop Area (+800 sq.ft surplus)
System Target Capacity	15.00 kW (Single LT Installation)
Total Installed Array Rating	15.40 kWp
Solar PV Module Rating	550W Mono PERC / TOPCon Half-Cut
Total Module Count	28 Nos.

[METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

Total Rooftop Area	2000.00 Sq.Feet (185.81 sq.m)
Utilized Usable Rooftop Area	2000 Sq.Feet (185.81 sq.m)
First Year Annual Generation	24,417.31 kWh
25-Year Cumulative Generation	542,500.49 kWh
Estimated Payback Period	4 Years & 5 Months
Post-Payback Profitability (25 Yrs)	Rs. 7,242,468.56 (INR 72.42 Lakhs)
Alternative FD 6.5% Benchmark (25 Yrs)	Rs. 3,167,372.91 (INR 31.67 Lakhs)

ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 444 Tonnes CO2 | Tree Equivalence: 710 Mature Trees | Green Energy: 100% Clean LT Power

Based on standard Central Electricity Authority (CEA) grid emission and forestry carbon sequestration benchmarks.

[ARRAY] Building 1 Layout & Rooftop Array Sizing

Building structural envelope, array capacity, and module layout

Building / Array Identification	Building 1 (Rooftop Array)
Gross Rooftop Area	2000.00 Sq.Feet (185.81 sq.m)
Utilized Effective Rooftop Area	2000 Sq.Feet (185.81 sq.m)
Building Plant Target Capacity	15.00 kW
Solar PV Module Rating	550 W Mono-PERC (Mono-facial)
Number of Solar Modules	28 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	11 kW, 415V 3-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (11 kW)

Core inverter rating (70% derated selection), MPPT topology & models

Selected Inverter Rating	11 kW Grid-Tied Inverter (Three-Phase)
Certified Top-Tier OEM Models	Deye 11K-G04, Kstar Solar
Derated AC Design Power (70% Factor)	10.78 kW AC (from 15.00 kWp DC)
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx16 Sq.mm
DC Array Isolator Switch	50A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	25 Amps
Rated Inverter Operating Current	15.00 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx16 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx6 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	20.0 Meters (ACDB to Utility Net-Meter)

[BOM] Turnkey Plant Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV modules	550W Mono-PERC (Mono-facial)	28 Nos.	Rs. 280,000
02	On-grid inverter	11 kW, three-phase (Deye 11K-G04 or equiv.)	1 No.	Rs. 117,333
03	Rooftop mounting structure	HDGI, module-compatible	1 Set	Rs. 70,000
04	DC solar cable	6 sq.mm copper, PV-rated	160 m	Rs. 13,600
05	AC cable	25 sq.mm copper	85 m	Rs. 29,750
06	MC4 connectors	PV-rated connector pairs	28 pairs	Rs. 8,400
07	DCDB	Isolator + fuse + Type-II DC SPD	1 No.	Rs. 14,000
08	ACDB	25A MCB + 40A RCCB + Type-II AC SPD	1 No.	Rs. 14,500
09	Earthing kit	Earth electrodes & accessories (IS 3043 - 3 Pits)	3 Nos.	Rs. 7,500
10	Lightning arrester	Standard rooftop PV type	1 No.	Rs. 6,000
11	Installation accessories	Lugs, ferrules, clamps, conduit, glands, etc.	1 Lot	Rs. 25,000
12	Installation & commissioning	Labour and commissioning	1 Job	Rs. 70,000
		TOTAL PLANT-SIDE CAPITAL COST (12-ITEM BOM)		**Rs. 656,083.33**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget

Official DISCOM statutory filing fees and net-metering charges

Application & Registration Fee	Rs. 1,000 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 2,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 11.8 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 12,000 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 8,000 (Refundable utility escrow)

TOTAL CAPITAL INVESTMENT REQUIRED

Turnkey Plant BOM Cost: Rs. 656,083 | ESCOM Utility Budget: Rs. 23,012 | Grand Total Outlay: Rs. 679,095 (Rs. 6.79 Lakhs)

[FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

CAPITAL INVESTMENT (CAPEX)	PAYBACK PERIOD (RETURN)	25Y NET SAVINGS (PROFIT)	FD 6.5% BENCHMARK (BENCHMARK)
Rs. 656,083	**4Y 5M**	**Rs. 72.42L**	**Rs. 31.67L**
12-Item BOM Basis	Break-Even Year	Rs. 7,242,469	Alternative Asset

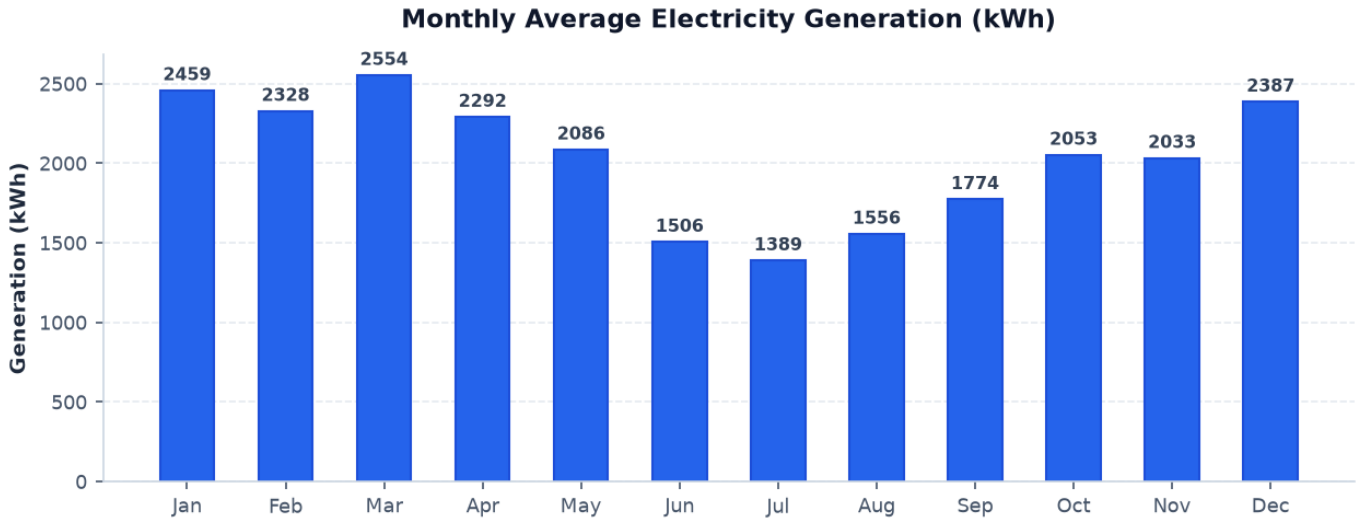
Yr	Gen (kWh)	Tariff (Rs/u)	Gross Rev (Rs)	AMC (Rs)	Net Annual (Rs)	Cumulative Net (Rs)
01	24,417	Rs. 8.05	196,559	9,000	187,559	187,559
02	24,173	Rs. 8.45	204,263	9,000	195,263	382,822
03	23,931	Rs. 8.87	212,272	9,000	203,272	586,094
04 *	**23,692**	**Rs. 9.32**	**220,810**	**9,000**	**211,810**	**797,904**
05	23,455	Rs. 9.78	229,392	9,000	220,392	1,018,296
06	23,221	Rs. 10.27	238,476	9,000	229,476	1,247,772
07	22,988	Rs. 10.78	247,815	9,000	238,815	1,486,587
08	22,759	Rs. 11.32	257,627	9,000	248,627	1,735,213
09	22,531	Rs. 11.89	267,893	9,000	258,893	1,994,106
10	22,306	Rs. 12.48	278,374	9,000	269,374	2,263,480
11	22,083	Rs. 13.11	289,503	9,000	280,503	2,543,983
12	21,862	Rs. 13.76	300,818	9,000	291,818	2,835,801
13	21,643	Rs. 14.45	312,743	9,000	303,743	3,139,544
14	21,427	Rs. 15.18	325,257	9,000	316,257	3,455,801
15	21,212	Rs. 15.93	337,914	9,000	328,914	3,784,715
16	21,000	Rs. 16.74	351,545	9,000	342,545	4,127,261
17	20,790	Rs. 17.57	365,286	9,000	356,286	4,483,546
18	20,582	Rs. 18.45	379,745	9,000	370,745	4,854,292
19	20,377	Rs. 19.37	394,694	9,000	385,694	5,239,986
20	20,173	Rs. 20.34	410,315	9,000	401,315	5,641,301
21	19,971	Rs. 21.36	426,582	9,000	417,582	6,058,884
22	19,771	Rs. 22.43	443,472	9,000	434,472	6,493,356
23	19,574	Rs. 23.55	460,960	9,000	451,960	6,945,315
24	19,378	Rs. 24.73	479,216	9,000	470,216	7,415,531
25	19,184	Rs. 25.96	498,020	9,000	489,020	7,904,552

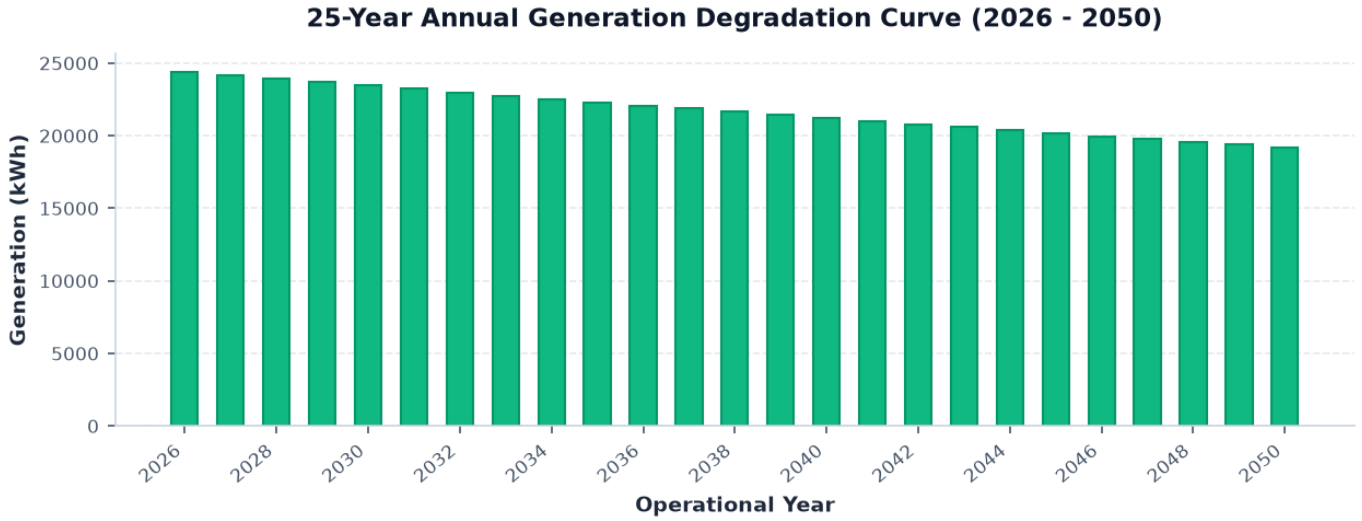
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

[ANALYTICS] Generation Analytics & Performance Curves

Monthly solar insolation yield and 25-year degradation profile

SPECIFIC YIELD (YIELD)	PERFORMANCE RATIO (PR)	CAPACITY UTIL. (CUF)	DEGRADATION RATE (DEGRADATION)
1628 kWh/kWp	**75.0%**	**18.1%**	**1.0% / year**
kWh/kWp/Year	IEC 61724 Standard	Annual Plant Factor	Tier 1 Linear Warranty





Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.

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ON-GRID SOLAR PV SYSTEM

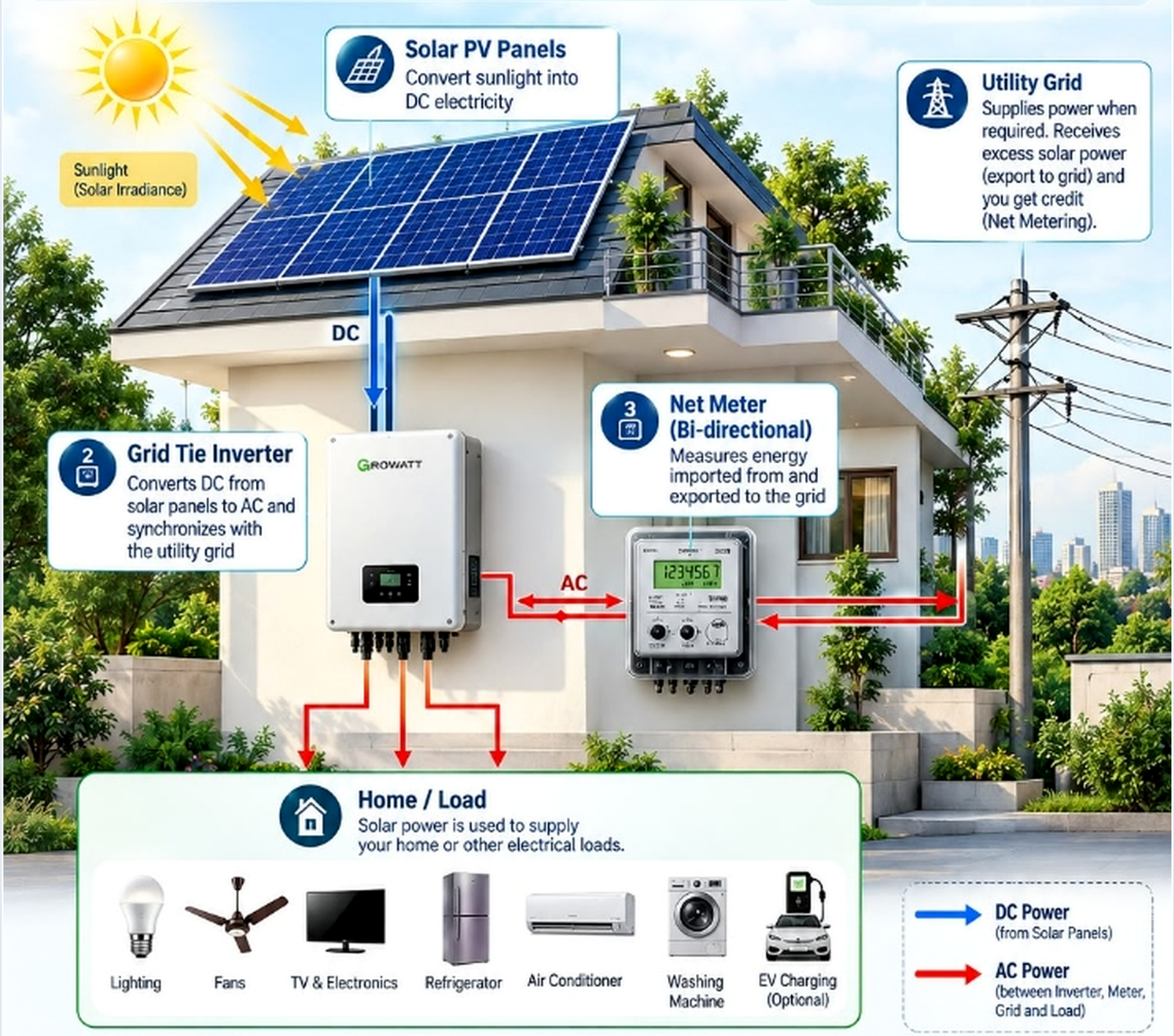
(Grid Connected Solar System)

GENERATE • CONSUME • EXPORT • SAVE

Harness The Sun For A Better Tomorrow

An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.

- Lower Electricity Bills
- Clean Energy
- Sustainable Future



Home / Load

Solar power is used to supply your home or other electrical loads.

Lighting | Fans | TV & Electronics | Refrigerator | Air Conditioner | Washing Machine | EV Charging (Optional)

DC Power (from Solar Panels)

AC Power (between Inverter, Meter, Grid and Load)

KEY ADVANTAGES

- Reduced Electricity Bills
- Lower Carbon Emissions
- No Battery Required
- Net Metering Benefits
- Low Maintenance
- Environment Friendly

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Clean Energy Better Tomorrow